Cantilever: HOLLOW DESIGN ACCEPTED

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vshell run summary: Cantilever
================================================================================

decision
--------
  outcome                         HOLLOW design accepted
  reason                          best hollow design changes stiffness by -0.8%
                                  and load capacity by +162.6% at constant mass
                                  (J = +0.4788)
  best design (k solid regions)   7
  final mesh                      final_design.stl

part
----
  STEP file                       /home/claude/vshell_repo/examples/step/Solid
                                  Cantilever.step
  volume (B-rep) mm3              6.102e+04
  volume (field) mm3              6.099e+04
  mass g                          63.46
  faces                           81
  analytic faces                  39
  support faces                   [10, 17, 25, 33]
  load faces                      [[42]]
  force N                         [0, 0, -200]

segmentation
------------
  joints                          4
  struts                          5
  mounts                          5
  rank order (most shear-critical first) [3, 9, 4, 1, 2, 6, 5, 8, 7]

baseline (solid)
----------------
  stiffness K0 N/mm               256.8
  load capacity P0 N              431.3
  governing mode                  Euler buckling
  lambda strength                 9.331
  lambda Euler                    2.157
  von Mises q MPa                 3.922
  mount von Mises q MPa           6.711
  FE DOF                          517686
  FE cells                        139840
  CG iterations                   34

best hollow candidate
---------------------
  solid regions                   [1, 2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 14]
  hollow regions                  [7, 8]
  outer offset / strut wall s mm 1.913
  as-built wall mm                1.713
  stiffness K N/mm                254.8
  K/K0                            0.9922
  load capacity P N               1133
  P/P0                            2.626
  J                               0.4788
  governing mode                  Euler buckling
  governing region                7
  lambda strength / Euler / local [9.105, 5.663, 94.66]
  mass (field) g                  63.43
  enclosed voids                  2
  filled small voids mm3          0
  mesh volume mm3                 6.089e+04
  mesh watertight                 no

search
------
  hollow designs analysed         9
  feasible in mass sweep          9
  objective                       {'w_stiffness': 0.5, 'w_capacity': 0.5,
                                  'min_gain': 0.02, 'floor_stiffness': 0.85,
                                  'floor_capacity': 0.85}

grid
----
  sdf_mm                          0.4
  shape                           [426, 376, 376]
  fe_mm                           0.8
  seg_mm                          1
  n_exact                         4571645
```

n_active	3661998

timings s

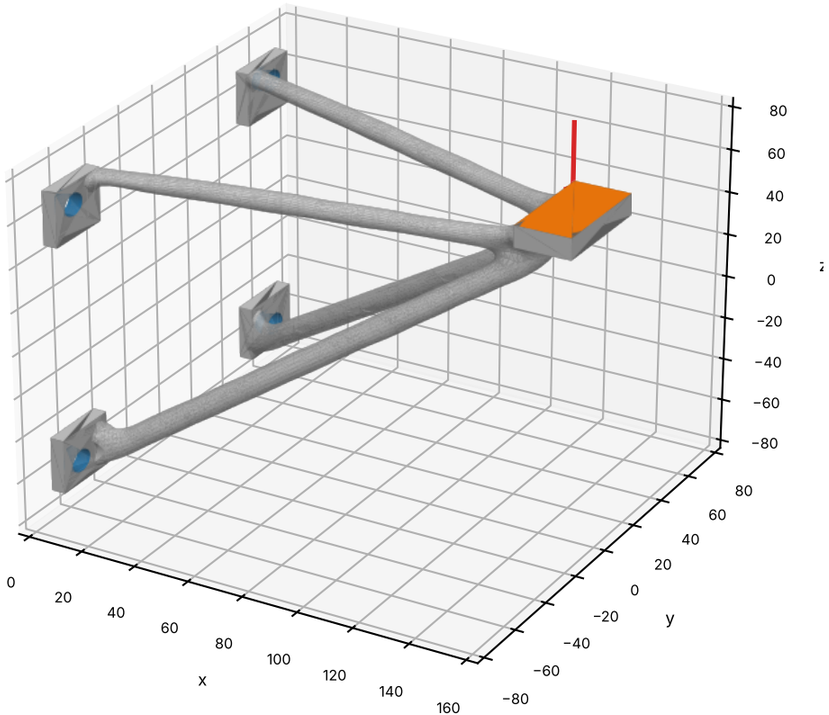
geometry	2.801
sdf	236.8
segmentation	12.16
fe	645.1
mass_sweep	83.03
outputs	9.319
total	1106

provenance

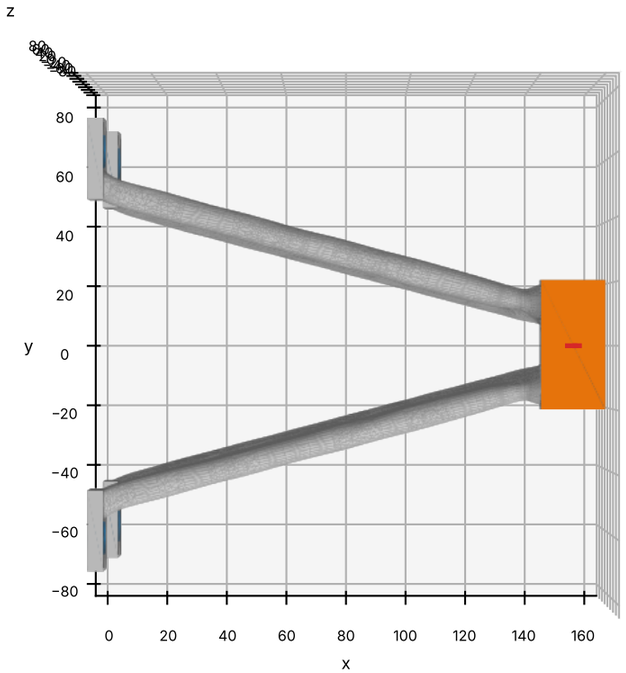
time (UTC)	2026-10-28T08:06:25+00:00
host	vm
platform	Linux-6.18.44-fc-v180-x86_64-with-glibc2.39
python	3.13.18
git	-
STEP sha256	6bc427af481b617f
packages	numpy 2.5.3, scipy 1.18.1, scikit-image 0.26.0, skan 0.13.1, libigl 2.6.3, pyamg 5.3.0, trimesh 5.2.0, matplotlib 3.11.2, build123d 0.13.0

Cantilever: supports (blue, all DOF fixed) and load (orange faces, red arrow, |F| = 200 N)

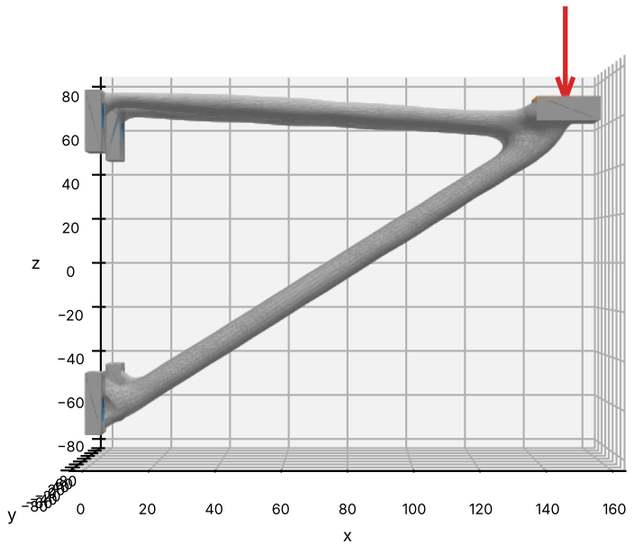
isometric



top (+z)

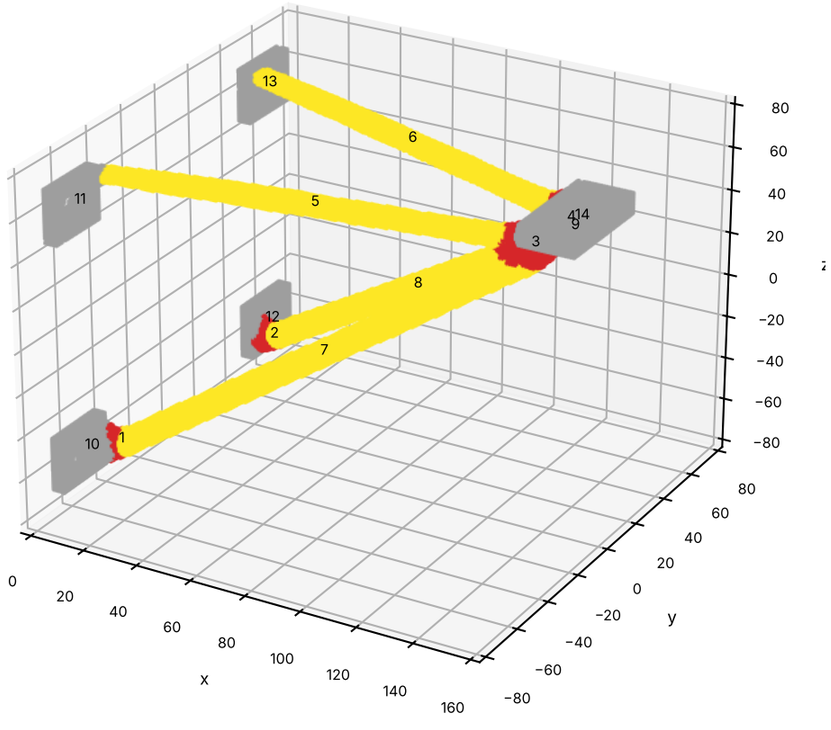


front (-y)

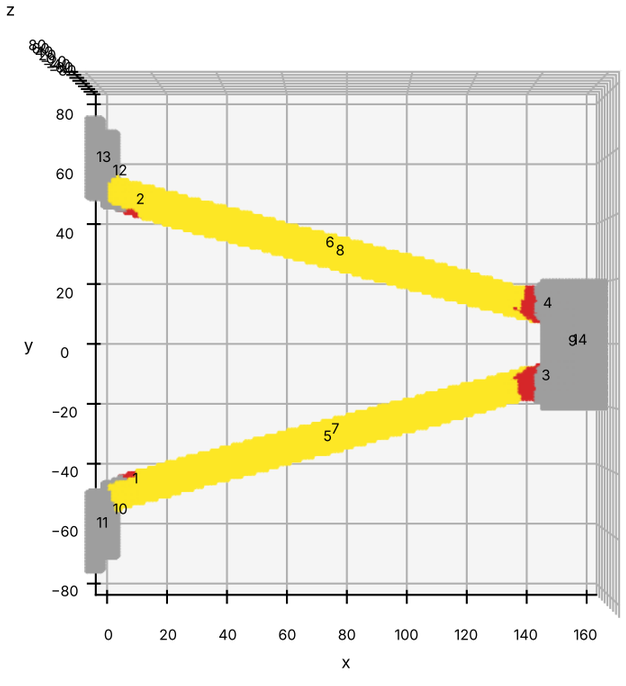


Segmentation: 4 joints (red), 5 struts (colour = slenderness lambda, 0 to 60+), 5 mounts (grey, never hollowed); numbers = region ids

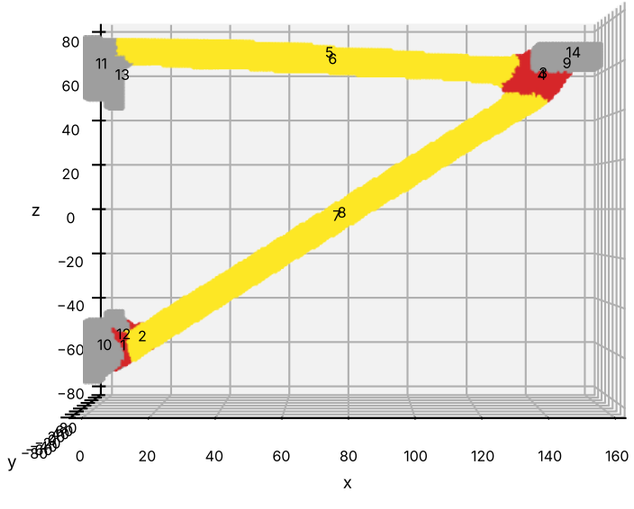
isometric



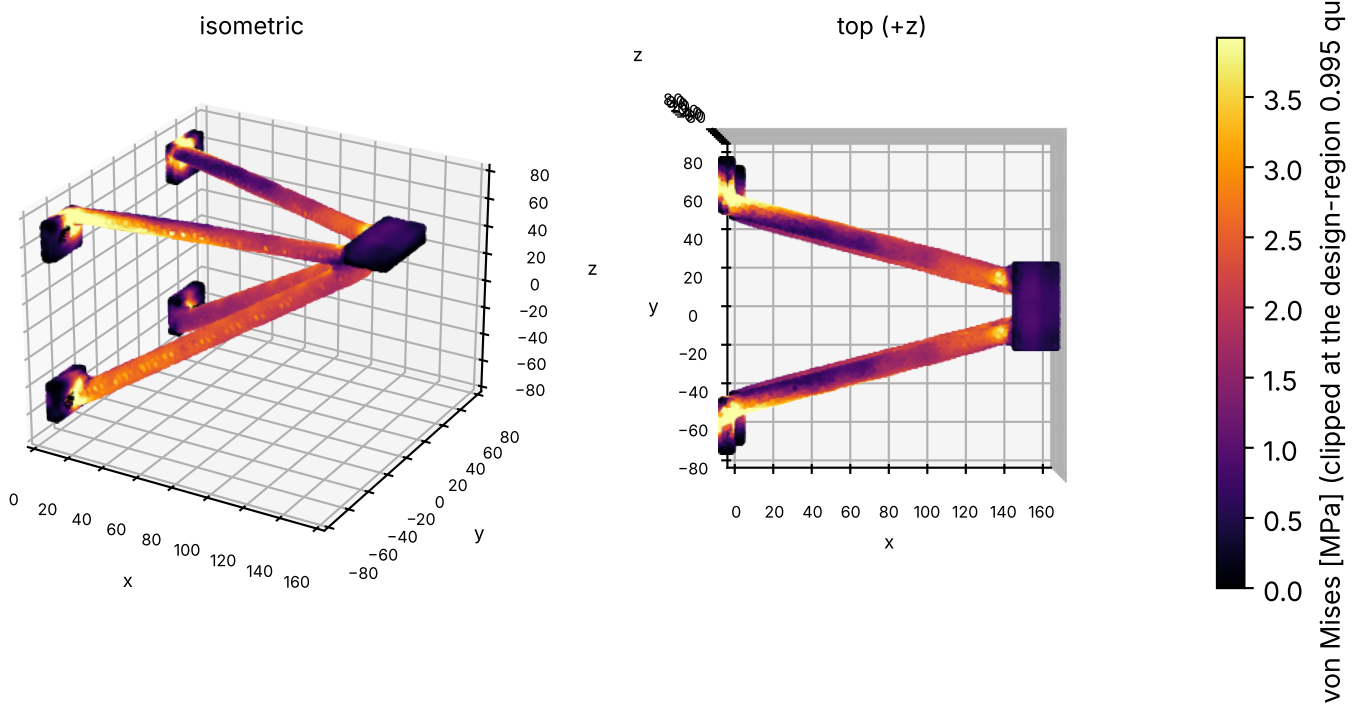
top (+z)

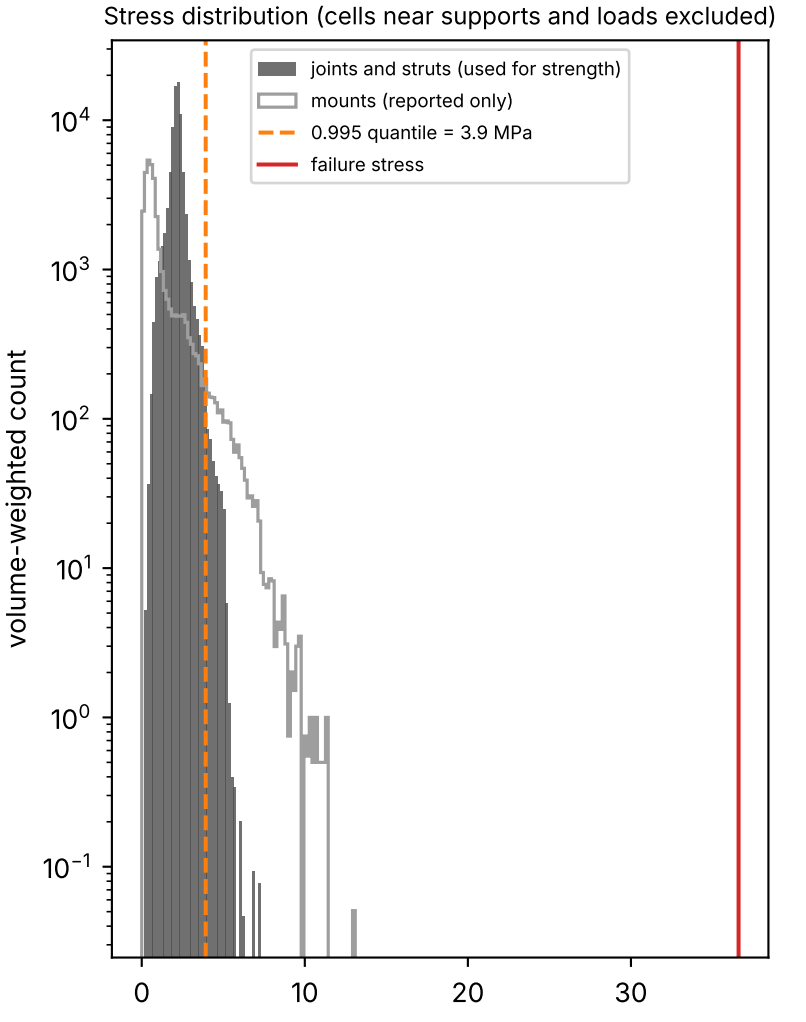


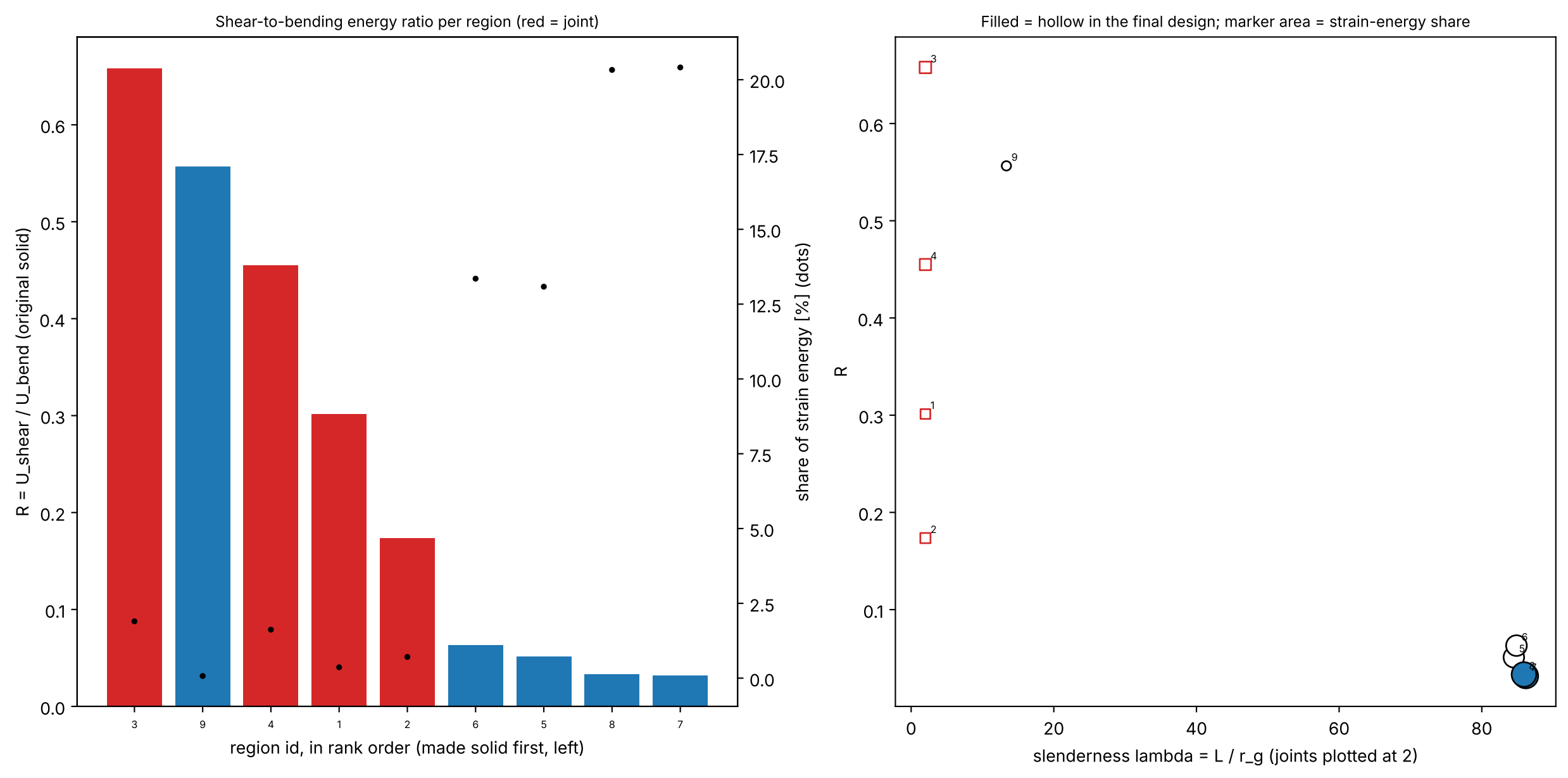
front (-y)

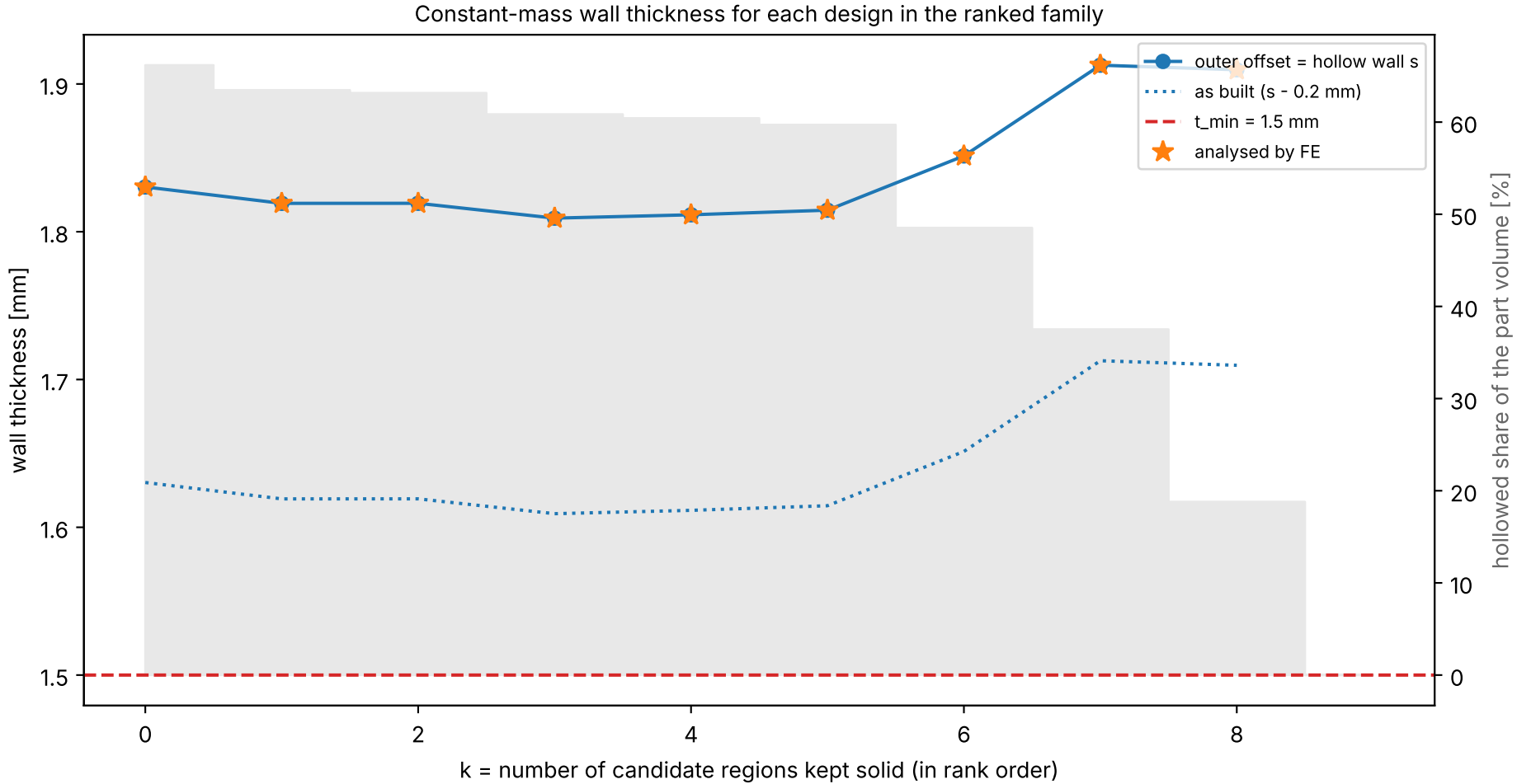


Original solid: von Mises stress at |F| = 200 N



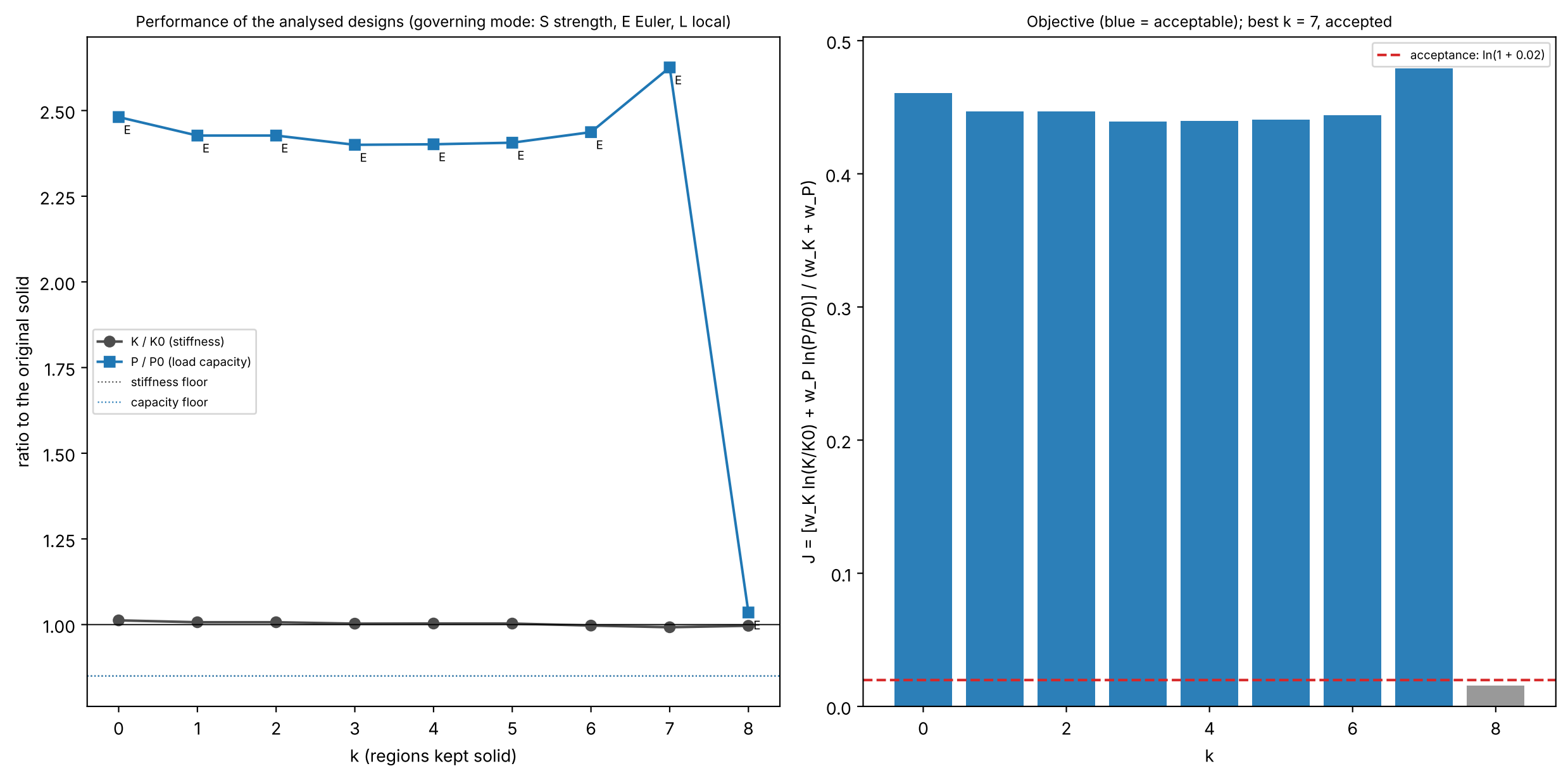


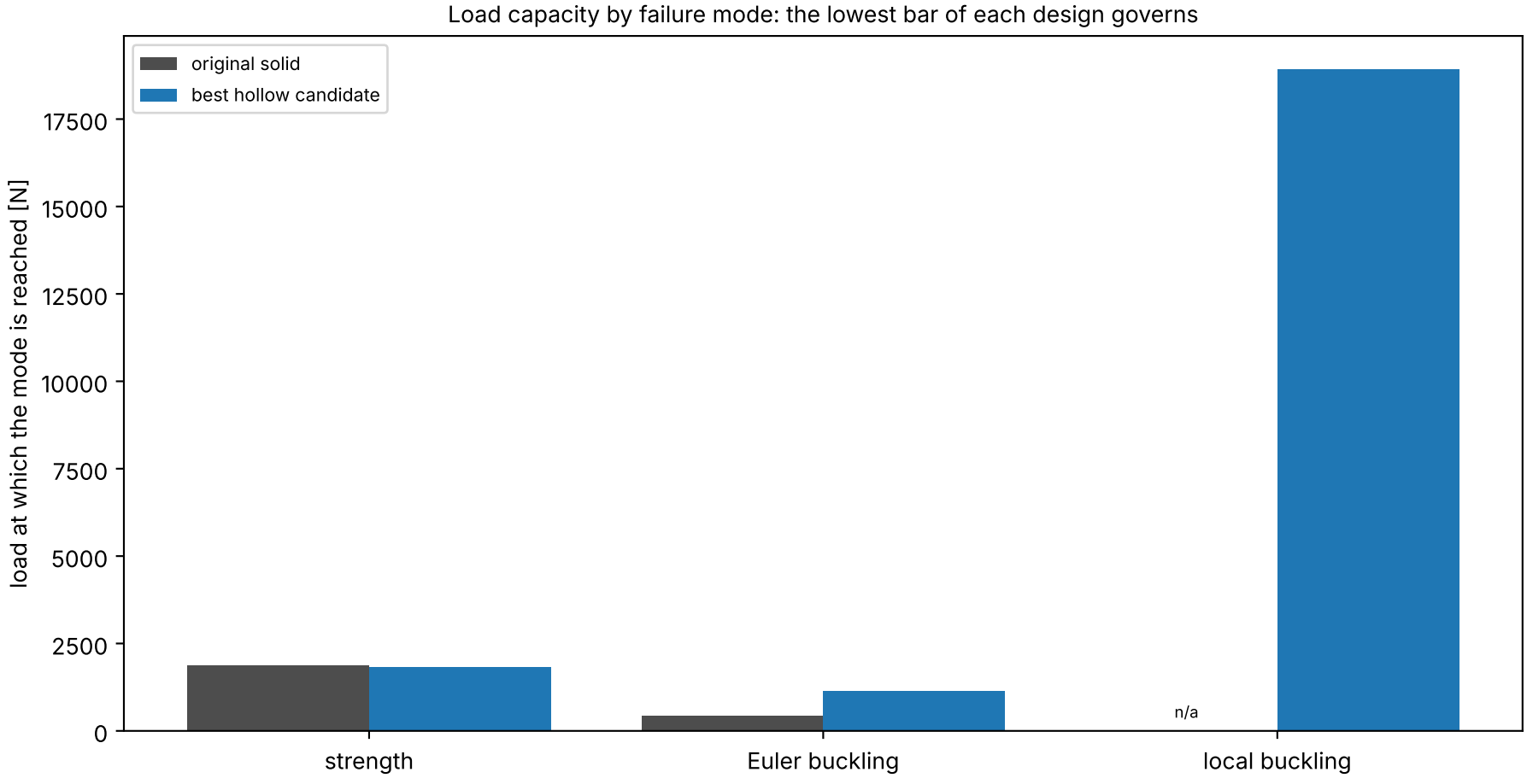

Shear-to-bending energy ratio per region (red = joint)
R = U_shear / U_bend (original solid)
share of strain energy [%] (dots)
region id, in rank order (made solid first, left)
3
9
4
1
2
6
5
8
7
Filled = hollow in the final design; marker area = strain-energy share
R
slenderness lambda = L / r_g (joints plotted at 2)

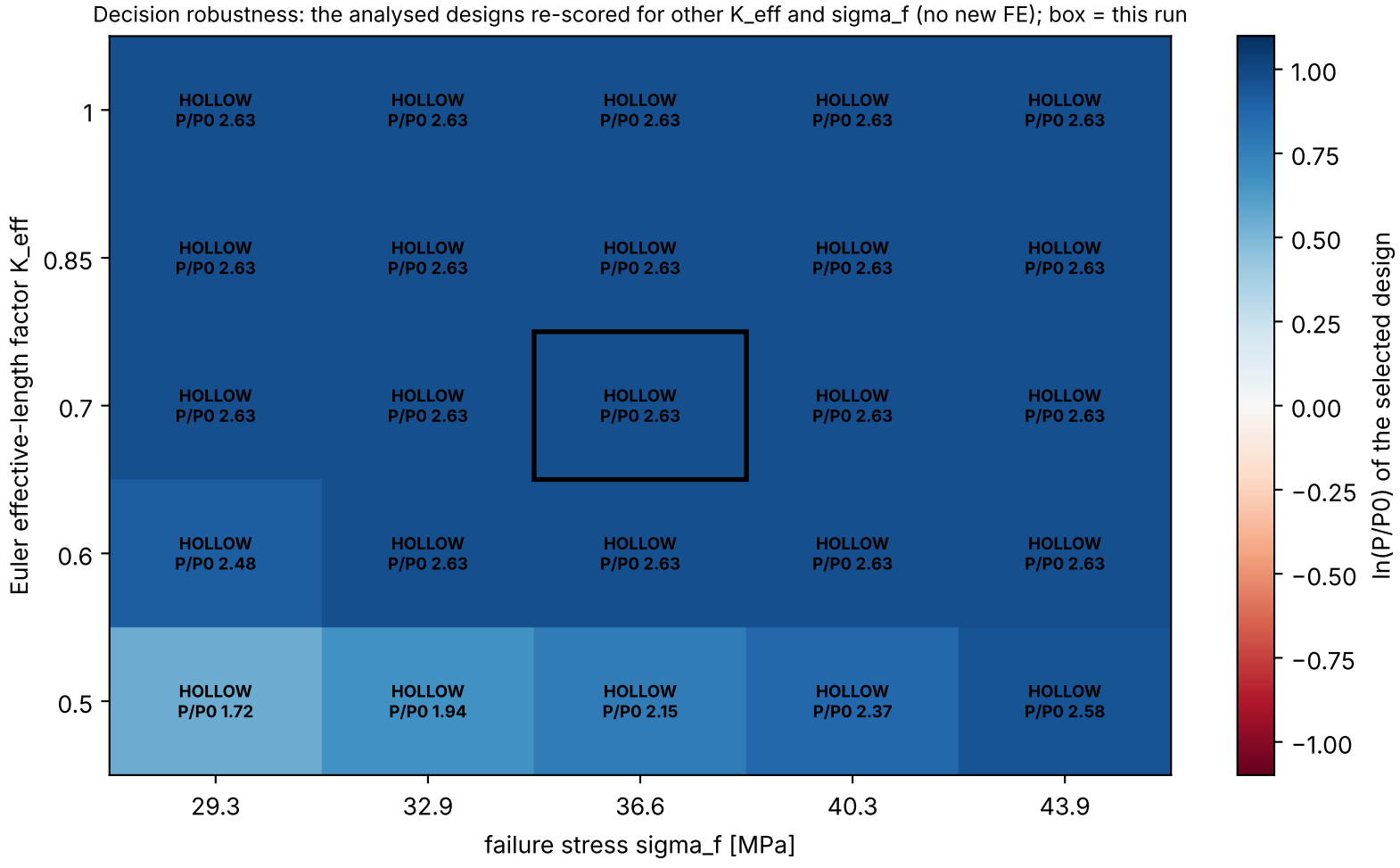

Constant-mass wall thickness for each design in the ranked family
outer offset = hollow wall s
as built (s - 0.2 mm)
t_min = 1.5 mm
analysed by FE
wall thickness [mm]
hollowed share of the part volume [%]
k = number of candidate regions kept solid (in rank order)
1.9
1.8
1.7
1.6
1.5
60
50
40
30
20
10
0
0
2
4
6
8

Regions (blue rows are hollow in the final design)

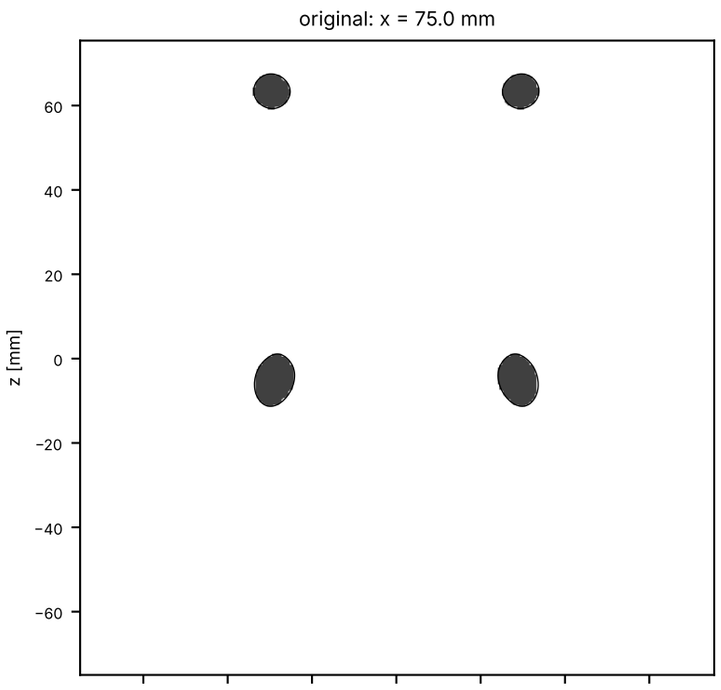
id	kind	rank	volume mm3	r mm	L mm	lambda	R	U share	final
1	joint	4	260	3.16	-	-	0.301	0.4%	solid
2	joint	5	434	4	-	-	0.174	0.7%	solid
3	joint	1	1.65e+03	3.61	-	-	0.658	1.9%	solid
4	joint	3	1.41e+03	3.16	-	-	0.455	1.6%	solid
5	strut	7	6.72e+03	3.61	152	84.5	0.051	13.1%	solid
6	strut	6	6.82e+03	3.61	153	84.9	0.063	13.4%	solid
7	strut	9	1.15e+04	4.36	188	86.2	0.0317	20.4%	hollow
8	strut	8	1.14e+04	4.36	187	85.9	0.0333	20.3%	hollow
9	strut	2	188	5.39	35.9	13.4	0.556	0.1%	solid
10	mount	-	3.24e+03	4.47	-	-	1.06	6.1%	solid
11	mount	-	2.91e+03	4	-	-	1.15	7.6%	solid
12	mount	-	3.22e+03	4.47	-	-	1.1	5.9%	solid
13	mount	-	2.9e+03	4	-	-	1.07	7.5%	solid
14	mount	-	8.19e+03	7	-	-	1.35	1.1%	solid

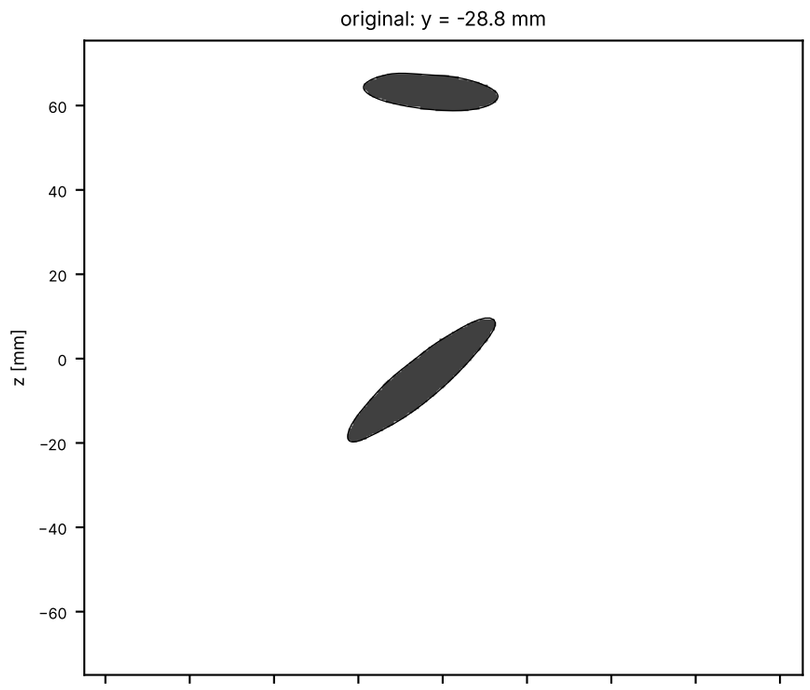

Performance of the analysed designs (governing mode: S strength, E Euler, L local)
ratio to the original solid
k (regions kept solid)
K / K0 (stiffness)
P / P0 (load capacity)
stiffness floor
capacity floor
E
Objective (blue = acceptable); best k = 7, accepted
J = [w_K ln(K/K0) + w_P ln(P/P0)] / (w_K + w_P)
k
acceptance: ln(1 + 0.02)

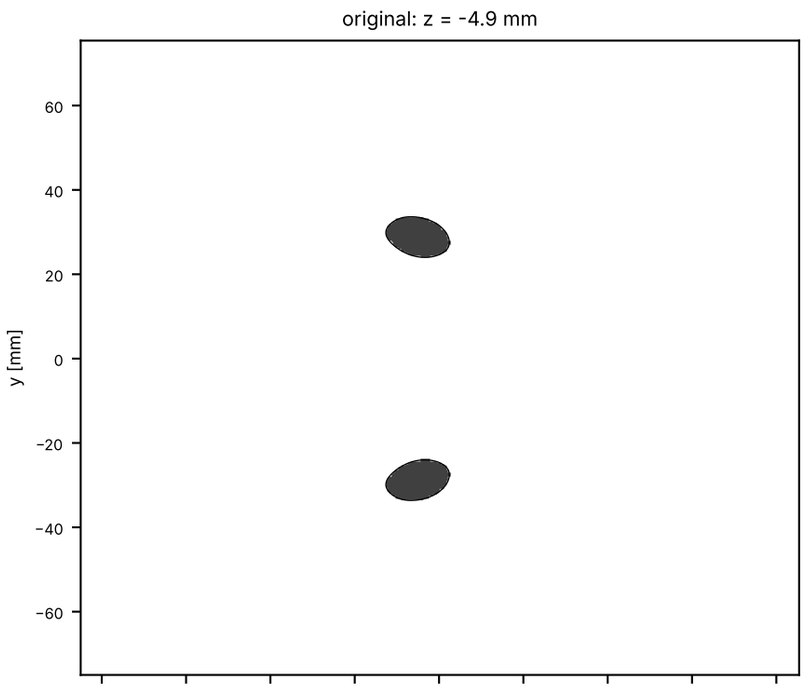

Load capacity by failure mode: the lowest bar of each design governs
original solid
best hollow candidate
load at which the mode is reached [N]
0
2500
5000
7500
10000
12500
15000
17500
n/a
strength
Euler buckling
local buckling

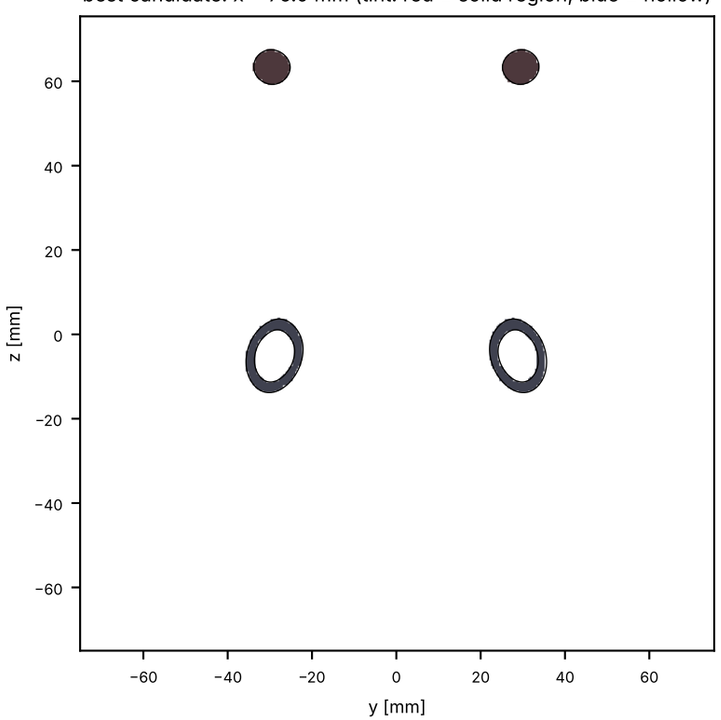

Decision robustness: the analysed designs re-scored for other K_eff and sigma_f (no new FE); box = this run
HOLLOW P/P0 2.63
HOLLOW P/P0 2.48
HOLLOW P/P0 1.72
HOLLOW P/P0 1.94
HOLLOW P/P0 2.15
HOLLOW P/P0 2.37
HOLLOW P/P0 2.58
Euler effective-length factor K_eff
1
0.85
0.7
0.6
0.5
29.3
32.9
36.6
40.3
43.9
failure stress sigma_f [MPa]
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
ln(P/P0) of the selected design

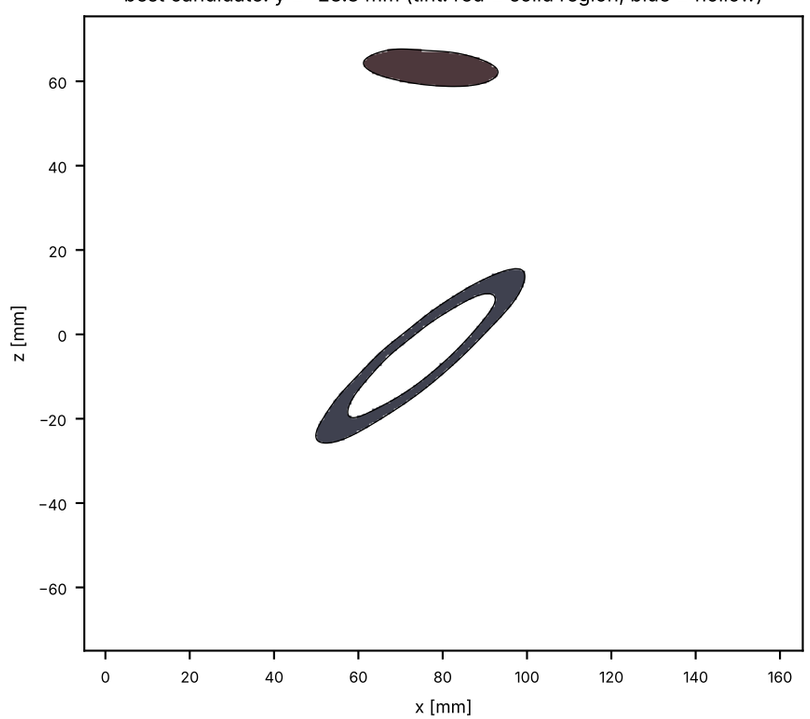
Sections through the largest hollow region of the best candidate (or the largest joint); material grey, void white

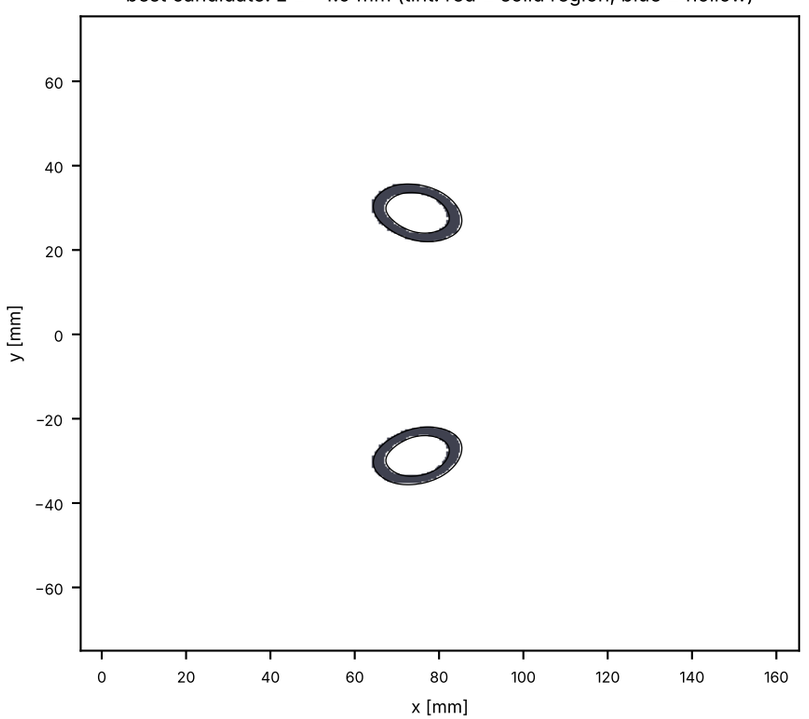




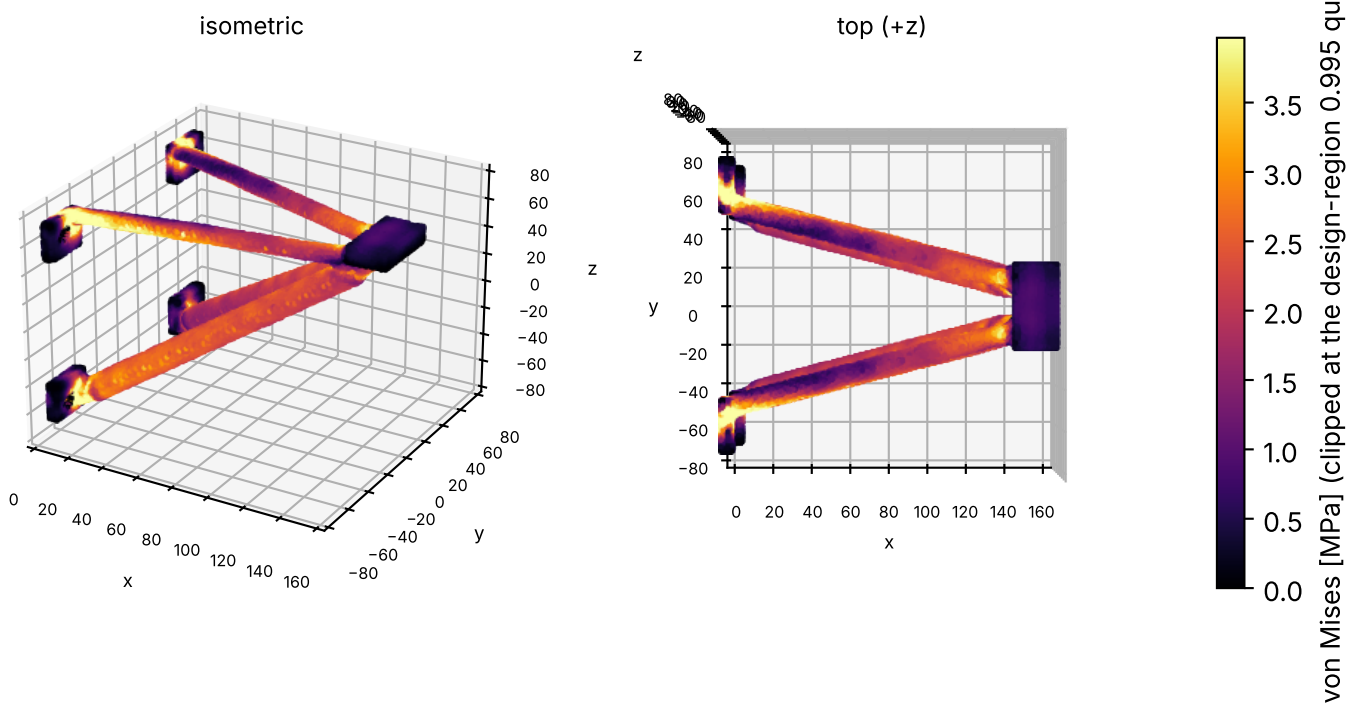


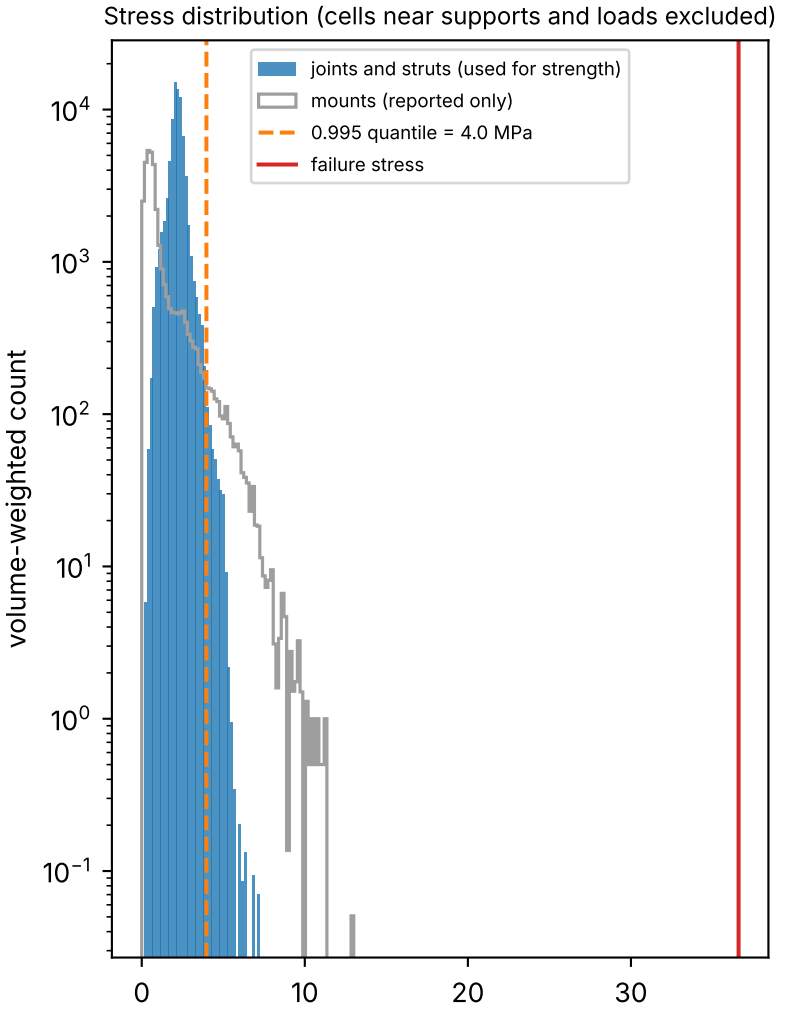






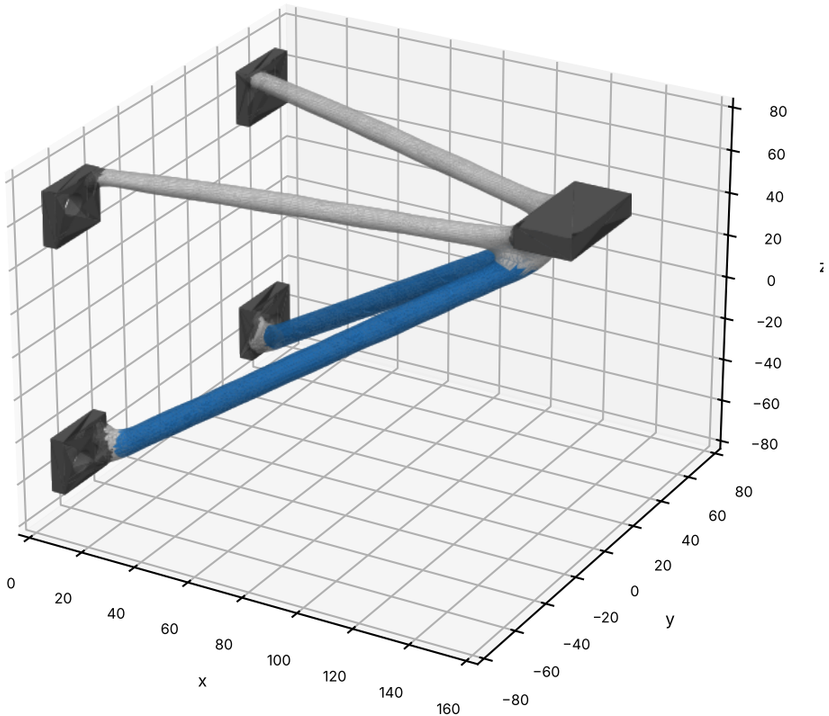
Best hollow candidate: von Mises stress at |F| = 200 N



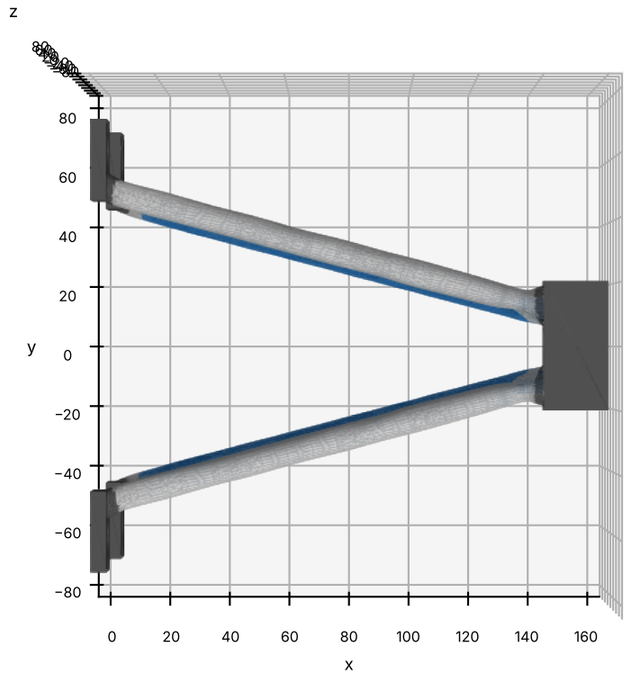


Best candidate: hollow regions blue (wall 1.91 mm), solid regions light grey, mounts dark grey

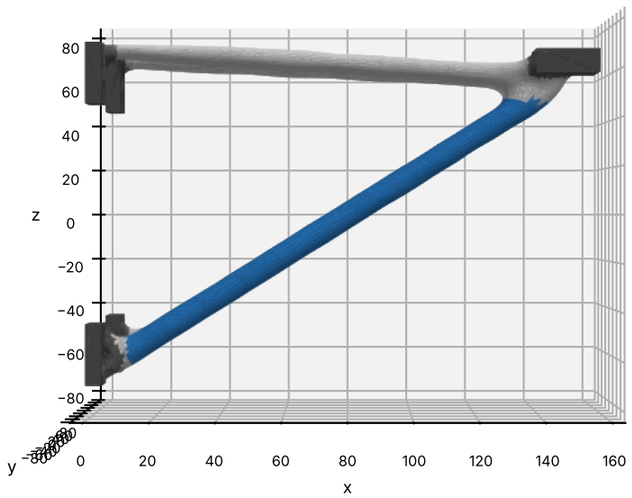
isometric



top (+z)



front (-y)

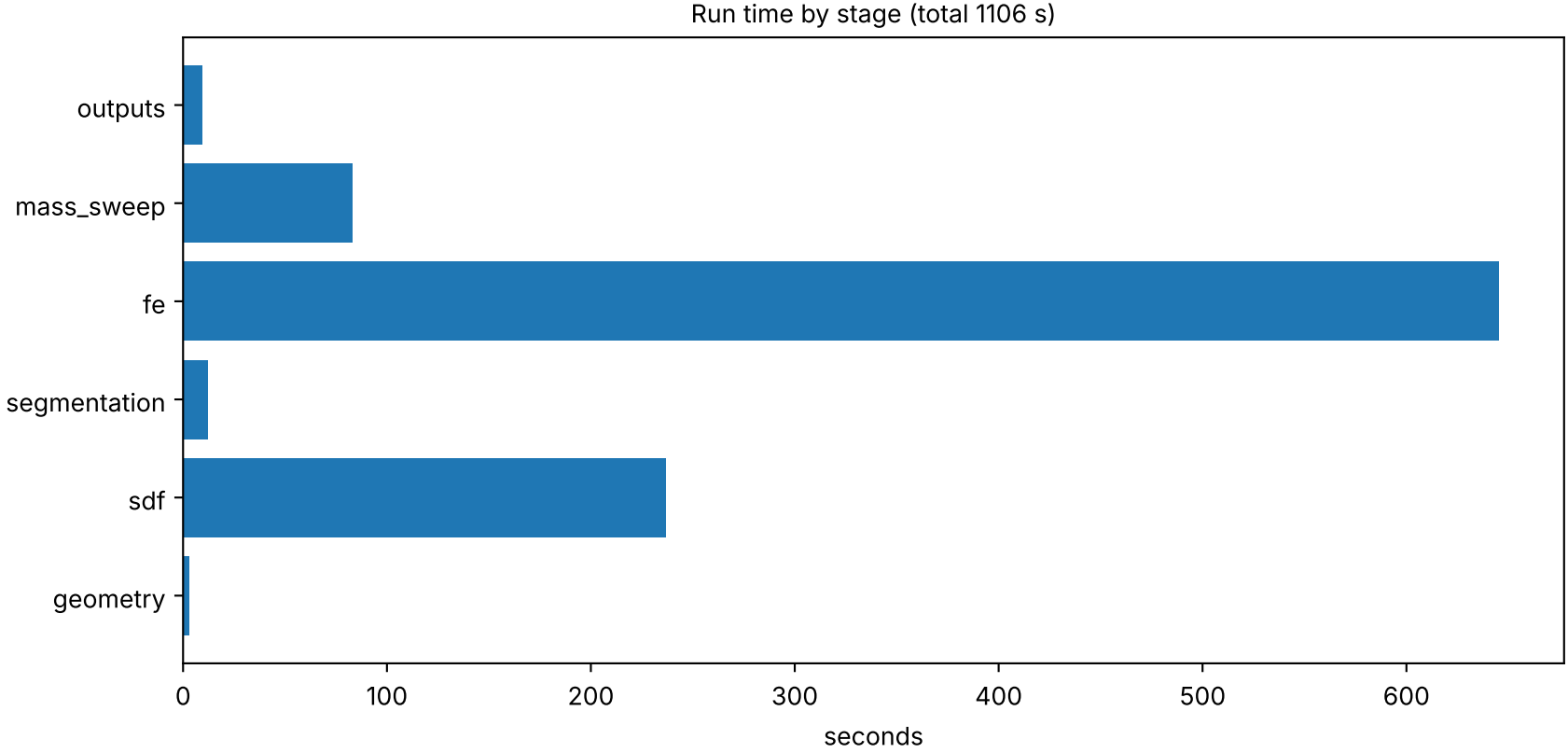


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Enclosed voids of the best candidate: 2
  void 2: 11180 mm3
  void 3: 11185 mm3

Small voids filled before analysis: 0.0 mm3

Suggested escape holes (lowest-stress wall points, outward direction n):
  void 2: (19.6, -39.6, -50.8) n = (-0.18, +0.97, -0.15), von Mises 1.5 MPa, d = 3.0 mm
  void 2: (31.2, -36.4, -41.2) n = (-0.11, +0.97, -0.19), von Mises 1.6 MPa, d = 3.0 mm
  void 3: (21.2, 39.2, -49.6) n = (-0.14, -0.98, -0.17), von Mises 1.5 MPa, d = 3.0 mm
  void 3: (36.0, 34.8, -36.8) n = (-0.18, -0.98, -0.12), von Mises 1.6 MPa, d = 3.0 mm

Escape holes are suggestions only: they are not cut into the mesh and their mass
(about pi d^2 t / 4 each) is not included in the constant-mass balance.
```


Run time by stage (total 1106 s)
outputs
mass_sweep
fe
segmentation
sdf
geometry
0
100
200
300
400
500
600
seconds